TrueTrip: Where to Stay, Given Where You Need to Go

Problem and audience

THE PROBLEM

Booking a hotel has two blind spots. The price leaves out parking and commute costs, and the rating is one platform's own score. Checking other platforms means more searches.

THE AUDIENCE

Travelers with fixed destinations, such as conference attendees and event-goers. Some rent a car, some ride. They watch their budget and compare several sites.

Illustrative example (made-up numbers): 3-night conference trip with a rental car

	Hotel A	Hotel B
Room + tax + fees	$600	$640
Parking (3 nights)	$180	$0
Travel time at $20/h	1.5 h = $30	3.0 h = $60
True trip cost	**$810**	**$700**
Page rating (10-pt)	8.7	8.4
Google rating (5-pt)	3.9	4.4

The insight

Hotel B costs $40 more in room rate but is $110 cheaper once parking and travel time count. Its page rating is lower, yet Google rates it higher: one platform's score can mislead.

Sorted by room rate, Hotel A wins. Sorted by true trip cost, Hotel B wins.

Proposed Solution

A browser extension that adds the costs and ratings hotel sites leave out

WHAT THE TRAVELER SEES ON BOOKING

Example Hotel — **8.7 Excellent** (3,634 reviews)

Per night $449 | $978 for 2 nights

ADDED BY TRUETRIP

Parking: $60/night x 2 = $120 (from the hotel's site)
Rides to your anchors: $50-74, about 1.1 h
Ratings: Google 4.1, Tripadvisor 4.0
True trip cost: about $1,182

Illustrative numbers; the hotel name is a placeholder.

WHAT THE TRAVELER DOES

- Searches on Booking as usual
- Sets once: 1-3 anchors, car or no car, limits
- Sees the TrueTrip block on every card
- Optional: sorts results by true trip cost
- Gets break-even notes, e.g. "A wins only if you skip the car"

$$C_{trip} = C_{room} + C_{tax} + C_{fees} + C_{parking} + C_{rides} + V \times T_{travel}$$

Parking applies if you drive, rides if you don't. V = your time value in $/hour (default $20).

Why this is different

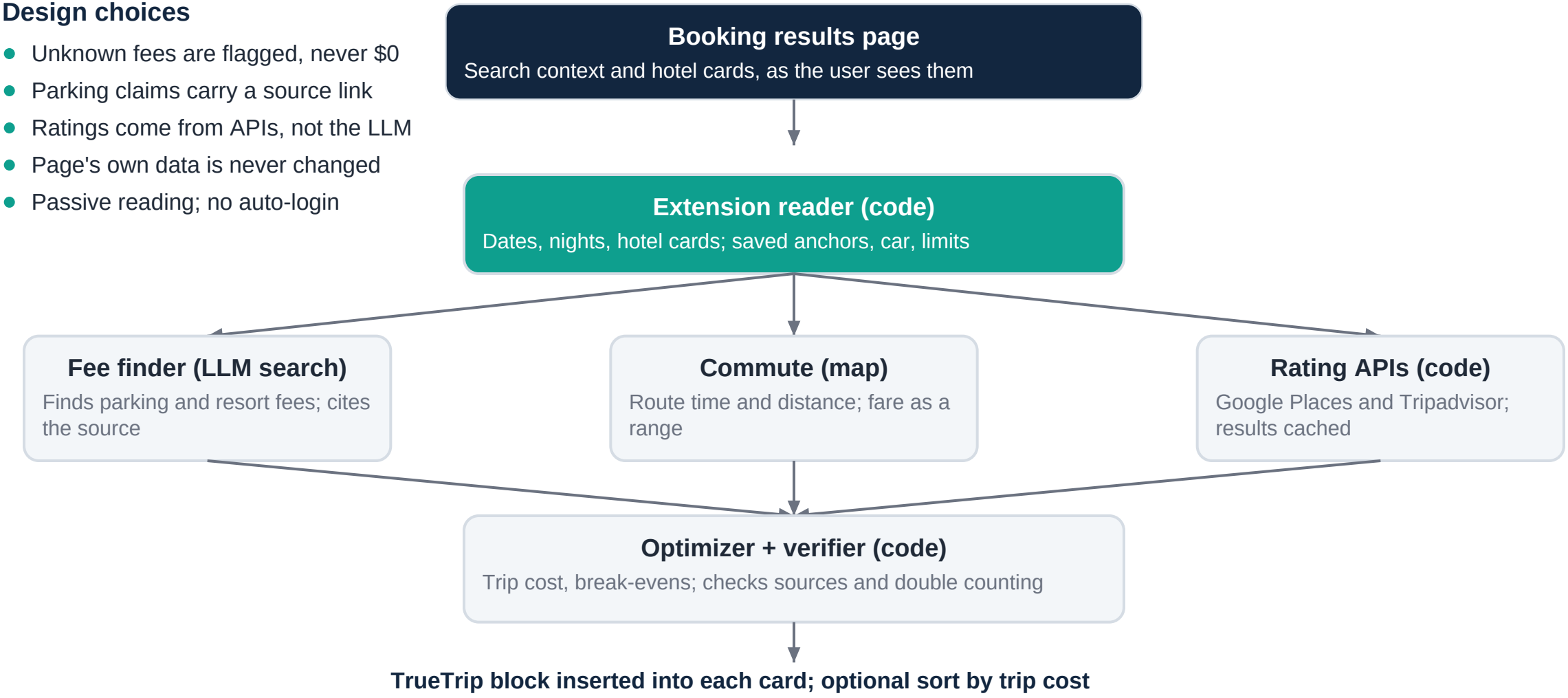
- Booking-style results show room price and one platform's rating; parking and commute are left to you
- TrueTrip adds parking (LLM search), commute (map) and cross-platform ratings to the page you already use
- Works inside Booking's own results page: no new site and no new search

Approach

Reads the page; LLM search for parking, a map for commute, APIs for ratings

Design choices

- Unknown fees are flagged, never $0
- Parking claims carry a source link
- Ratings come from APIs, not the LLM
- Page's own data is never changed
- Passive reading; no auto-login



Scope and Risks

Main risks: hotel data access, LLM-found fees, and matching hotels across sites

IN SCOPE FOR THE MVP

- Chrome extension for Booking.com results
- US hotels in any city, 1-3 anchors
- Drive (parking) or ride (fare range)
- Extra info on each card; optional re-sort
- Ratings: Google Places and Tripadvisor

Out of scope (later)

- Flight search and hotel booking
- Rental-car pricing, fuel, tolls, itineraries
- Other hotel sites; arrival-time check

KEY RISKS AND MITIGATIONS

- **No hotel API access**
  Extension reads the page the user has open
- **Site terms and page changes**
  Passive reading only; one site adapter; check terms first
- **LLM search misreads parking**
  Show the source; check a sample by hand; flag unknowns
- **Ride prices vary**
  Show a range; wider outside tested cities
- **Same hotel, different names**
  Match by name and location; check matches by hand

Decision: the first version only adds information to Booking's own results. It does not search or book, so effort goes into cost accuracy.

Plan and Success Criteria

Five milestones; accuracy checked by hand on a sample, plus a live demo

SUCCESS CRITERIA (about 30 hotels, checked by hand)

- Parking: source found for 70%+; fee correct for 80%+ of those
- Ratings: right hotel matched for 90%+
- Commute: fare range contains a real app quote for 70%+
- Safety: no fee shown as $0 without a source

HOW WE DEMONSTRATE IT

- Live demo on a real Booking results page
- Extra info appears on each card, with sources
- Worked cases, including a Price Reversal Case: lower room rate, higher true trip cost
- Accuracy results from the hand-checked sample

Milestones

1. Extension + reader
Reads cards, adds a block

2. Commute + cost
Map, C_trip, limits

3. Parking finder
LLM search + sources

4. Rating APIs
Google, Tripadvisor

5. Test + demo
Hand check, live run